

QUIZ

Theory of Automata (Section B)

Name: _____

Roll No: _____

Q1. For the alphabet $\Sigma = \{a, b\}$, what is the length of the string $w = abbaa$?

- A. 2
- B. 4
- C. 5
- D. Infinite

Q2. In a DFA, if you are in a state and receive an input, how many possible next states can you move to?

- A. Exactly one
- B. At least two
- C. Zero
- D. Any number

Q3. Which statement about NFA and DFA is correct?

- A. Every NFA has fewer states than its equivalent DFA.
- B. NFAs are strictly more powerful than DFAs.
- C. For every NFA, there exists an equivalent DFA.
- D. Some regular languages cannot be represented by DFAs.

Q4. A state in a Finite Automaton that has no outgoing transitions is often called a:

- A. Start state
- B. Final state
- C. Dead or Trap state

D. Transition state

Q5. The language accepted by a DFA is unchanged if:

- A. Final states are removed.
- B. Equivalent states are merged.
- C. Transitions are reordered.
- D. The alphabet size is reduced.

Q6. A Transition Graph (TG) differs from a DFA mainly because: A. It has no start state.

- B. It may have multiple start states and labels can be strings.
- C. It does not accept any language.
- D. It cannot have cycles.

Q7. Which of the following best describes a Transition Graph?

- A. A labeled directed graph without accepting states.
- B. A DFA without determinism constraints.
- C. A directed graph where edges are labeled with strings from Σ^* .
- D. An undirected graph with weighted edges.

Q8. In a Finite Automaton, the symbol λ (or ϵ) represents:

- A. An error
- B. The end of a string
- C. An empty string (no character)
- D. A space character

Q9. In formal diagrams, the final states are traditionally represented as:

- A. Nodes with no outgoing edges.
- B. Nodes with double circles.

C. Nodes with dashed borders.

D. Nodes with arrows pointing to them.

Q10. A Transition Graph with exactly one start state and exactly one outgoing transition per symbol from each state is:

A. An NFA

B. A PDA

C. A DFA

D. A Turing Machine

SECTION B: DESCRIPTIVE QUESTIONS (10 MARKS)

Answer all questions. (5 Marks Each)

Q11. Design a DFA over the alphabet $\Sigma = \{a, b\}$ that accepts all strings that start with 'a' and end with 'b'.

1. Draw the state transition diagram.
2. List the set of states (Q) and the final states (F).
3. Show the Transition Table.

Q12. NFA vs. Transition Graph: 1. Explain the difference between an NFA and a Transition Graph (TG). Mention at least two key features that a TG has that a standard NFA does not. (3 Marks)

2. Draw a small Transition Graph that accepts strings containing the word "**cat**" as a single transition label. (2 Marks)